

MULTIMODAL CYBERBULLYING DETECTION USING INTER-MODAL-ATTENTION-BASED DEEP LEARNING

ABSTRACT

Cyberbullying on social media poses significant psychological and social challenges, necessitating automated detection systems that can handle multimodal content. This work presents a novel multimodal cyberbullying detection framework that jointly analyzes textual and visual inputs using intermodal attention mechanisms. Textual embeddings are generated with BERT, while images are processed through BLIP-based captioning to capture contextual visual features. The model fuses these modalities via an inter-modal attention module and employs BiLSTM layers for classification. Additional preprocessing and postprocessing modules perform input validation, NSFW filtering, severity scoring, and category assignment, producing structured outputs suitable for real-time moderation. Evaluated on the MMHS150K dataset, the framework achieves 87.2% accuracy, 85.6% precision, 84.9% recall, and 85.2% F1-score, outperforming baseline methods including SAFE, MCNN, and LBP-sim by 4–7%. The proposed system offers a lightweight, interpretable, and deployment-ready solution, demonstrating effective detection of cyberbullying in complex multimodal contexts such as memes, sarcasm, and code-switched text.